Kinetic energy and net work

antocciphysics | TEACHER KEY | CPA

Learning target: Connect net work to a change in kinetic energy.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 1; 3 course days later.*
Factor 9, because $K=\frac{1}{2}mv^2$.

02. Explain a worked example (worked)
Bars 4 and 9 J; +5 J net work. Speed is not increased by 5 m/s.

03. Look and explain (model)
+6 J; speed increases.

04. Build the solution (practice)
9 J; final 16 J.

05. Use the idea (reason)
Negative friction work reduces network and final kinetic energy.

06. Choose a method (mixed) *Review: Unit 3, Day 6; 12 course days later.*
Yes: sticking defines perfectly inelastic. Total momentum is conserved; kinetic energy need not be.

07. Exit check (exit)
-12 J; speed falls.

08. STRETCH: Apply it another way (stretch)
Changes 4-1=3 J and 9-4=5 J; equal speed increases require different energy changes.

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01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 1; 3 course days later.*
K becomes 1/3 of the original; fixed p means K∝1/m.

02. Explain a worked example (worked)
Bars 4 and 9 J; +5 J net work. Speed is not increased by 5 m/s.

03. Represent and interpret (model)
+6 J; speed increases.

04. Solve and justify (practice)
Initial 8 J; final 32 J; v=√(64/4)=4 m/s.

05. Analyze the reasoning (reason)
Negative friction work reduces network and final kinetic energy.

06. Choose a method (mixed) *Review: Unit 3, Day 6; 12 course days later.*
A stops; B moves at +6 m/s. Equal masses exchange velocities in an ideal 1 D elastic collision.

07. Exit check (exit)
Final 4 J-initial 36 J=-32 J.

08. STRETCH: Apply it another way (stretch)
Initial 24 J, final 6 J, v=2 m/s; stopping work -24 J.